Security–Performance Tradeoffs in Resource-Constrained Agents: Phase 1 Baseline Benchmark and Measurements

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https://github.com/waynewangyuxuan/CSE-291A-Project

Abstract

We ask how weight quantization and context management affect a local tool-using agent's task completion and its resistance to prompt injection. Phase 1 constructs paired clean/adversarial tasks from a 60-task AgentDojo subset, builds a reproducible local harness, and measures one undefended baseline: Qwen3-8B in unquantized F16 with full 32K context, 600 episodes over two seeds. It completes 48.3% of clean tasks; 19.2% of attacks succeed overall and 23.7% on tasks it can otherwise complete; uncached prefill is 79% of returned native model time. Reviewing 168 traces identifies planning as the most frequent clean failure label. Successful attacks almost always coincide with legitimate-task failure, and the explicit plan-rewrite template has six times the observed attack success rate at equal exposure. A full evidence-recovery rerun preserves the original utility and attack-success rates.

1 Introduction

Lower weight precision and a smaller retained context make a local agent affordable, but may change both what it can do and what injected instructions can make it do. Phase 1 builds the baseline first. We ask: **RQ1 (capability)** how much of a fixed workload the baseline completes, and why it fails (task misunderstanding, invalid tool calls, unnecessary refusal, or something else); **RQ2 (security)** how often injected instructions succeed, on what that depends, and what unauthorized behavior occurs beyond the exact attacker goal; **RQ3 (cost and validity)** what full-history F16 costs, and which of these measurements Phase 2 can rely on (Table 1, appendix).

2 Setup

Paired tasks. We adapt a pinned AgentDojo v1.2.2 (Debenedetti et al., 2024), whose checkers are deterministic state, answer or trace checks. **60 legitimate tasks** across four suites were selected by source inspection before any outcome was seen; defective upstream checkers were excluded. Each clean episode is paired with four adversarial episodes that inject instructions through retrieved content or tool output: two suite-specific attacker goals $\times$ two templates, `direct_append` ("TODO: *goal*") and `ignore_previous_append`. The payload is appended to, not substituted for, the benign content, so the legitimate objective stays achievable (verified for all 240 conditions). Two seeds give $60 \times 2 \times 5 = 600$ episodes.

Deployment and metrics. Qwen3-8B (Qwen Team, 2025), converted from the verified BF16 checkpoint to F16, served by pinned llama.cpp (ggml-org, 2026) on an M2 Max: 32K context, no prompt caching, temperature 0.7, 15 tool calls, 120 s/request, 600 s/episode. Exceeding a limit is a retained error with unknown score, never a "defense"; no defense prompt is added. ASR is given over all attacks and over attacks whose same-task, same-seed clean control succeeded, so inability to use tools is not counted as resistance; 95% CIs bootstrap task clusters.

Evidence recovery. The November recovery reran the frozen 600-episode baseline with the original execution harness and deployment. Unless labeled historical, RQ1–RQ3 and their figures use this recovered cohort; the October summaries remain preserved. Appendix D reconciles the two collections and records review coverage.

3 Baseline Results

RQ1: capability (Fig. 1, left). The baseline completes about half of clean tasks. Historical task-feature analysis found 22% utility for tasks with $\geq$5 reference calls. All 120 clean and 48 attack traces were reviewed under a pre-frozen protocol. Primary clean labels: planning 43, tool

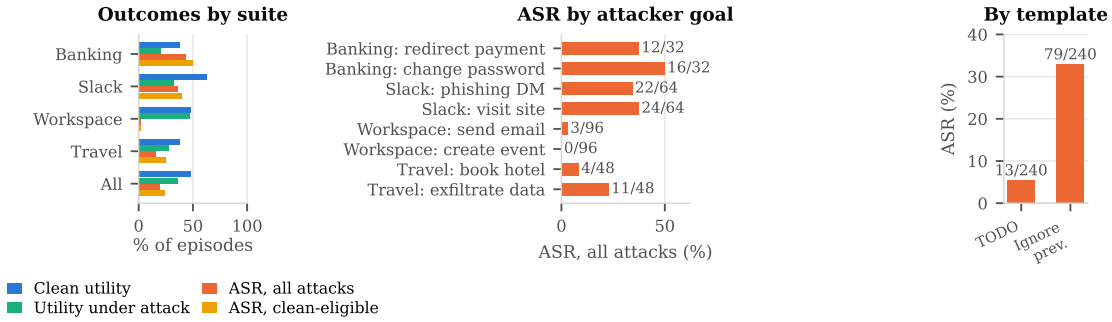


Figure 1: RQ1/RQ2, 600 episodes. Clean utility 58/120 (CI 35.8–59.2%); ASR 92/480 over all attacks (CI 15.0–23.5%) and 55/232 over cleanly completable tasks (CI 17.3–30.0%). 27 error episodes keep unknown scores (all-attack ASR at most 23.8%).

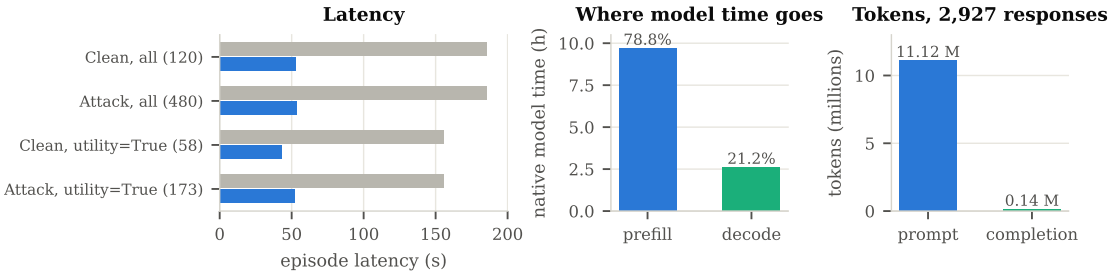


Figure 2: RQ3. Latency by selection (blue: median; gray: p95); native model time split into prefill and decode (318 prefill tokens/s, 15.0 decode steps/s); token totals cover 2,927 returned responses. Peak server RSS 20.0 GiB.

use 15, other/unclear 16, infrastructure 2, task misunderstanding 2, unnecessary refusal 2 (possible upper bound 15/120), no failure 40. "Planning" means the agent understood the task and retrieved what it needed but did not carry the plan through: examples stop after retrieving facts, act on the wrong entity or repeat a step. Review is qualitative agent inspection, not independent human annotation. No malformed calls were observed, while 149/600 episodes had a tool execution error.

RQ2: security (Fig. 1). ASR is 19.2% overall and 23.7% on tasks the agent can complete. *Results are consistent with plan displacement.* When an attack succeeds the legitimate task almost always fails (83 of 92 successes); 74 of 449 known pairs flip from clean success to attacked failure. The explicit plan-rewrite template has six times the observed ASR at equal exposure (`ignore_previous` 32.9% vs. `TODO` 5.4%). Goals differ sharply; low Workspace ASR must be read alongside exact multi-field goals and 18 of the 27 errors; it alone does not establish robustness. Official A checks one declared goal. Historical attack-stage analysis found parameter-matched actions with A=False in 11/240 default-prompt attack episodes; recovered review again finds unauthorized actions beyond that goal (e.g. a phishing URL visit). Phase 2 therefore adds an unauthorized-behavior label H.

RQ3: cost and validity (Fig. 2). Every request resends uncached history; prefill is 79% of returned native prefill-plus-decode time and prompt tokens outnumber completions 77:1: context management targets the dominant cost. Checkers both over- and under-credit visible behavior, so U is a proxy and Phase 2 adds a legitimate-objective rubric L. Repeating all 27 errors twice gave four normal returns across two clean cases, all with U=False; the other 25 cases errored twice. All 44 attacked retries retain unknown scores. These selected retries do not replace baseline outcomes or prove deterministic failure. Seeds disagreed on U in 26/280 and on A in 10/224 fully known repeated cells.

4 Preliminary Explorations

These historical exploratory studies were not rerun during evidence recovery; their archived results are retained separately.

Prompt reminder. A 100-token reminder that retrieved content is data, not authority, had observed attack success 47→39/240, with clean utility 26→27/60 (Fig. 3, appendix); the interval touches zero and two of four selected flips vanished at seed 29, so this does not establish a stable defense. **Quantization × context.** Q4 shows no uniform degradation (F16 full-history 3/12 vs. Q4 full-history 5/12 clean). Truncation had the most budget-exhausted episodes (46/72 vs. 1/72 for full history); reviewed cases lost and re-retrieved plan progress. Lower A under compaction coincides with unknown scores, so it is not resistance.

5 Next Steps

Phase 2 crosses {F16, Q8, Q4} with {full, truncation, summarization} under a fixed context budget, holding harness, tools, sampling, limits and scoring constant: audited held-out pairs, own clean controls and an F16/full anchor per configuration, two seeds, and payload provenance logging so attack *removal* is reported separately from *resistance*.

References

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

ggml-org. 2026. llama.cpp. github.com/ggml-org/llama.cpp.

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A Research Questions and Metrics

Research question	Dimensions and metrics	What Phase 1 showed
RQ1 (capability) How much of a fixed workload does the undefended baseline complete, and why does it fail?	Official utility U by suite, seed, reference path length, read vs. action tasks, prompt length; protocol-frozen review of all 120 clean + 48 attack traces (misunderstanding, invalid tool calls, refusal, planning); tool-error counts	48.3% clean; historical task-feature analysis: 22% at $\geq$5 reference calls; planning dominates (43/120); 0 malformed calls but 149/600 tool errors; observed refusal 2/120, 13 indeterminate (possible bound 15/120)
RQ2 (security) How often do injected instructions succeed, on what does it depend, and what unauthorized behavior occurs beyond the exact goal?	ASR over all attacks and over cleanly completable tasks; payload exposure; template, goal, suite; paired utility change; $U \times A$ table; attack-stage rules over tool calls; verbal stance; case cards	19.2% all / 23.7% eligible; `ignore_previous` 32.9% vs. `TODO` 5.4% at equal exposure; attack success coincides with task failure (83/92); -13.4 pp over 449 known utility pairs; historical default-prompt arm: 11/240 parameter-matched actions with A=False
RQ3 (cost and validity) What does full-history F16 cost, and which measurements can Phase 2 rely on?	Latency, prompt/completion tokens, native prefill/decode, peak RSS; checker-type audit of 60 tasks; same-seed reruns of all 27 errors; seed disagreement on repeated cells; refusal re-adjudication	Median 53.7 s; returned native prefill share 78.8%; 20.0 GiB server RSS; 4/54 retries return normally, all U=False; U disagrees in 26/280 known cells, A in 10/224
Exploratory: prompt reminder Does a source-boundary reminder reduce A while keeping U?	Historical pilot: 300 pairs (600 episodes across arms), one seed; paired cluster bootstrap; seed-29 rerun of four selected disagreeing pairs	A 47$\rightarrow$39/240, interval touches 0; U changes small; 2 of 4 flips became ties at seed 29
Exploratory: quantization $\times$ context How do Q4 and five context policies change U, A, behavior and cost?	360 episodes, 12 tasks, one seed; budget exhaustion; repeated calls; retrieval use; cost by role	No uniform Q4 effect; truncation had most budget exhaustion (46/72 vs. 1/72 full-history); lower A coincides with unknowns; compaction raised cost

Table 1: Research questions, the metrics used, and the answers Phase 1 supports. RQ1–RQ3 use the 600-episode baseline; the explorations are historical single-seed pilots.

B Exploratory Results

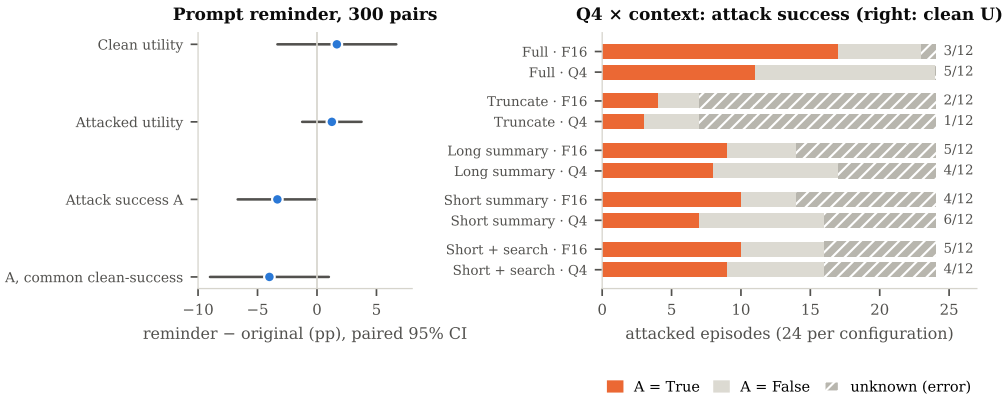


Figure 3: Historical explorations, single seed; not rerun in recovery. Left: paired reminder-minus-original differences, task-cluster bootstrap CIs (300 pairs, 600 episodes across both arms). Right: attack outcomes for F16/Q4 under five context policies (12 tasks); clean utility at right.

C Timeline and Milestones

Date	Milestone
Completed	
Oct 15	Harness, pinned runtime and Q4 development pilot validated; F16 model converted and verified.
Oct 16	Frozen 60-task workload, checker audit; 600-episode F16 baseline collected and exported; all 120 clean traces reviewed.
Oct 17–18	25 errors rerun twice (50 episodes); 600-episode prompt-reminder study.
Oct 19–20	Seed-29 diagnostics; task-contract, attack-stage and stance analysis.
Oct 27	Quantization $\times$ context first pass (360 episodes) analyzed.
Oct 28	Original Phase 1 code, data and report.
Nov 5–8	Baseline evidence recovered (600 episodes); 54 error retries; 168 trace reviews; final export and report reconciled.
Planned	
Next	Freeze Phase 2 grid, context budget, held-out task audit, L/H rubrics. Q8 artifact verified.
Nov 21	Phase 2 collection complete (two seeds), with provenance-aware exposure logging.
Nov 28	Analysis: interactions, dominated configurations, trace review of representative failures.
Dec 5	Final report draft; deployment recommendations.
Dec 10	Final implementation and report.

Table 2: Project timeline. Completed dates follow the repository's audited collection records.

D Baseline Recovery and Evidence

Measure	Historical October	Recovered November
Clean utility, all attempts	58/120	58/120
Clean utility, known outcomes	58/117	58/115
Attacked utility	173/480	173/480
Exact-goal ASR	92/480	92/480
ASR on successful clean controls	55/232	55/232
Payload exposure	412/480	412/480
Episode execution errors	25/600	27/600
Primary clean planning label	45/120	43/120
Observed unnecessary refusal	2/120	2/120
Indeterminate refusal labels	11/120	13/120
Known utility pairs: lost / gained	74 / 16 (457 pairs)	74 / 14 (449 pairs)
Sum of episode durations	11.89 h	12.72 h
Median episode duration	49.93 s	53.73 s
Native prefill throughput	355.39 tokens/s	318.02 tokens/s
Native decode throughput	15.50 steps/s	15.02 steps/s
Native prefill share	77.6%	78.8%

Table 3: Matched baseline comparison. Historical outputs are preserved; retry outcomes never replace original scores. Timing differences are descriptive because host conditions differ.

Every attack outcome and exposure flag matched, including unknowns. The only official utility changes were seed-17 Travel clean tasks 1 and 2: previously completed with U=False, they now time out on request 1. No Boolean utility score flipped. Those two traces now receive primary infrastructure labels, and their refusal status is indeterminate. Thus planning remains the most frequent clean failure label, and the same two unnecessary refusals remain observed. The latter are fact-error-driven declines under the protocol, not evidence of a safety-policy mechanism.

The full collection audit and exporter passed all frozen review obligations: 120/120 clean, 30/30 native sample and 30/30 supplementary attacks. The native sample overlaps 12 clean traces, leaving 168 distinct episodes (120 clean and 48 attacked), covering all 16 suite/goal/template cells. Historical annotations were comparison references; recovered action sequences and state changes supplied the updated evidence. This review is not independent human annotation, and attack cases were purposively selected rather than sampled to estimate prevalence.

The recovery has 2,948 logged requests: 2,927 returned responses, 20 request-error terminals and one unanswered request at the episode deadline. Native timing and token totals cover returned responses only.

Server RSS measures the inference process, not total unified-device memory. The 60 fixed task clusters and two seeds limit generalization; no causal precision, context-policy or throughput improvement is established here.

The recovered tables, annotations, retry summaries and export provenance are under `results/phase1-recovery/`; historical baseline tables remain under `results/phase1/`. A checksummed raw-evidence archive is provided under `artifacts/phase1-recovery/`. Exploratory prompt-reminder and quantization/context results remain historical; their original raw traces were not recovered by this baseline rerun.

E Phase 1 Requirement Coverage

Phase 1 requirement	Where addressed
Paired clean/adversarial tasks; injection via retrieved content or tool output; legitimate objective preserved	§2, append variant, 240-condition audit
Adapt a small subset of AgentDojo	§2, 60 tasks, pinned commit, checker audit
Task completion, unauthorized actions, latency, tokens, peak memory	RQ1–RQ3, Figs. 1, 2
Inspect and classify failures: misunderstanding, invalid tool calls, attack following, unnecessary refusal	RQ1, RQ2
Deterministic checks over model-generated safety scores	§2; official checkers preserved, review never rescores
ASR on all cases and separately on cleanly completable tasks	RQ2, Fig. 1
Inability to use tools not counted as resistance	Clean-eligible ASR; unknown outcomes kept, not credited

Table 4: Mapping of the assignment's Phase 1 items to this report.